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Inventor(s)

QIAN; Jiale et al.

CRYSTALLINE FORM OF ELACESTRANT AND PREPARATION METHOD AND USE THEREOF

Abstract

Provided are novel crystalline forms of Elacestrant (Referred to as “Compound I”) and preparation methods thereof, pharmaceutical compositions containing the crystalline form, and uses of the crystalline form for preparing estrogen receptor depressant drugs and drugs for treating breast cancer with ER+, HER2-, ESR1 mutations.

##STR00001##

Inventors:	QIAN; Jiale (Suzhou, CN), SHI; Jiaming (Suzhou, CN)
Applicant:	CRYSTAL PHARMACEUTICAL (SUZHOU) CO., LTD. (Suzhou, Jiangsu, CN)
Family ID:	1000008669970
Assignee:	CRYSTAL PHARMACEUTICAL (SUZHOU) CO., LTD. (Suzhou, Jiangsu, CN)
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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure pertains to the field of chemical crystallography, particularly relates to crystalline forms of Elacestrant, preparation method and use thereof.

BACKGROUND

[0002] Breast cancer is cancer that develops from breast tissue. When breast cancer cells test estrogen receptor-positive, it's called estrogen receptor-positive (ER-positive) breast cancer. HER2 (human epidermal growth factor receptor 2) are receptors that sit on the surface of breast cells. When a breast cancer is HER2-negative, it means that the cancerous cells do not contain high levels of HER2. At present, breast cancer is the most common cancer in the world, and its incidence rate ranks first among female malignant tumors. More than 2 million people are diagnosed every year in the world.

[0003] Surprisingly, a new generation of oral estrogen receptor degrading agent (SERD) Elacestrant, jointly developed by Radius Health and Menarini Group, has been approved in the United States. It is for the treatment of postmenopausal women or adult men with advanced or metastatic breast cancer patients with ER+, HER2-, ESR1 mutations who have previously received at least first-line endocrine therapy. Elacestrant has good selectivity towards estrogen receptors and has anti estrogen effects in different tissues.

[0004] The chemical name of Elacestrant is: (R)-6-{2-{ethyl [4-(2-ethylaminoethyl)benzyl]amino}-4-methoxyphenyl}-5,6,7,8-tetrahydronaphthalen-2-ol, (Referred to as Compound I), and the structure is shown as the follows:

##STR00002##

[0005] It is well known in the field that drug polymorphism is a common phenomenon in small molecule drug development and it is an important factor affecting drug quality. A crystalline form is a solid material whose constituents are arranged in a highly ordered microscopic structure, forming a crystal lattice that extends in all directions. Polymorphism refers to the phenomenon that a compound exists in more than one crystalline form. Compounds may exist in one or more crystalline forms, but their existence and characteristics cannot be predicted with any certainty. Different crystalline forms of drug substances have different physicochemical properties, which can affect drug's in vivo dissolution and absorption and will further affect drug's clinical efficacy and safety to some extent. In particular, for some poorly soluble oral solid or semi-solid dosage forms, crystalline forms can be crucial to the performance of drug product. In addition, the physiochemical properties of a crystalline form are very important to the manufacturing process. Therefore, polymorphism is an important part of drug research and drug quality control.

[0006] Therefore, in order to obtain crystalline forms with acceptable physicochemical properties (including chemical stability, thermal stability, solubility, hygroscopicity and/or particle size), and manufacturability (including yield, impurity rejection during crystallization, filtration properties, drying properties and milling properties), as well as formulation feasibility (including stability with respect to pressure or compression forces during tableting), it requires comprehensive research on the crystallization behavior of Compound I dihydrochloride to obtain crystalline forms that meets the pharmaceutical needs of Compound I.

[0007] At present, the prior art WO2018129419A1 disclosed three crystalline forms of Compound I

dihydrochloride, Form 1, Form 2 and Form 3. The prior art WO2020010216A1 further disclosed Form 1B of compound I dihydrochloride (hereinafter referred to as Form 1B), and the conversion relationship between Form 1, Form 2 and Form 3. FIG. 1 in WO2020010216A1 shows that Form 1 transitions to Form 3 under high humidity conditions, Form 2 transitions to Form 3 when humidity is below 40% RH, and Form 3 transitions to Form 2 when humidity is above 40% RH. It indicates that the humidity stability of Form 2 and Form 3 is poor, which is not conducive to quality control during the storage, transportation and production of raw materials and formulations. Even the relatively stable Form 1 is unstable under high humidity conditions. In order to overcome the problems of Form 1, Form 2 and Form 3, WO2020010216A1 further disclosed Form 1B. Table 17 in WO2020010216A1 disclosed that Form 1B can maintain stability at humidity above 90% RH. It indicates that Form 1B has good stability and is the crystalline form with the best properties and most suitable for drug development among the crystalline forms disclosed in the prior art.

[0008] Furthermore, step 6 of scheme 1 in WO2020167855A1 disclosed a preparation method for a crystalline form of stable Compound I dihydrochloride, but no characterization data of the crystalline form is disclosed. The inventors of the present disclosure have repeated the preparation method to obtain the crystalline form, and it was confirmed to be Form 1B in the prior art WO2020010216A1.

[0009] Based on the analysis of the prior arts mentioned above, it can be concluded that Form 1B is the crystalline form with the best properties. However, prior art Form 1B has some disadvantages that are not conducive to medicinal use. As disclosed in Table 17 of WO2020010216A1, Form 1B is a very small flaky particle mainly composed of aggregates. The particle size distribution of easily agglomerated crystalline forms are uneven, which may lead to uneven content of drug formulations, thus increasing the variability of in vitro dissolution.

[0010] Furthermore, after in-depth research by the inventor of this application, it was found that Form 1B still has disadvantages such as low solubility, poor flowability and compressibility, and high adhesiveness. Low solubility is not conducive to the absorption of drugs in vivo, which in turn leads to low bioavailability. Poor flowability and compressibility, as well as high adhesiveness, can result in uneven dispersion of raw materials and excipients during the mixing process. During the product tableting process, it is easy to occur molding failure, cracking and fragmentation, resulting in different contents of active ingredients in different drug particles in the same batch, which in turn leads to an increase in the unqualified rate of drug production. Furthermore, poor content uniformity or excessive weight difference will lead to fluctuations in the intake of active ingredients in different patients or at different time points of taking medicine in the same patient, resulting in changes in absorption and blood concentration, thereby affecting the efficacy and even causing toxicity.

[0011] In order to overcome the disadvantages of prior arts, a new crystalline form meeting the medicinal standard is still needed for the development of drugs containing Compound I dihydrochloride. The inventors of the present disclosure surprisingly discovered crystalline form of Compound I dihydrochloride, which have advantages in at least one aspect of solubility, hygroscopicity, purification ability, stability, adhesiveness, compressibility, flowability, in vitro and in vivo dissolution, and bioavailability, etc. In particular, the crystalline form of the Compound I dihydrochloride of the present disclosure not only has good stability but also has advantages such as better solubility, purification ability, flowability, compressibility, adhesiveness, and uniform particle size distribution, which solves the problems existing in the prior arts and is of great significance for the development of drugs containing Compound I.

SUMMARY

[0012] The present disclosure is to provide novel crystalline form of Compound I dihydrochloride, preparation method and pharmaceutical compositions comprising the novel crystalline form.

[0013] According to the objective of the present disclosure, crystalline form CSII of Compound I dihydrochloride is provided (hereinafter referred to as Form CSII).

[0014] In one aspect provided herein, the X-ray powder diffraction pattern of Form CSII comprises characteristic peaks at 2theta values of $10.3^{\circ}\pm 0.2^{\circ}$, $12.1^{\circ}\pm 0.2^{\circ}$ and $15.4^{\circ}\pm 0.2^{\circ}$ using CuK α radiation.

[0015] Furthermore, the X-ray powder diffraction pattern of Form CSII comprises one or two or three or four characteristic peaks at 2theta values of $7.9^{\circ}\pm 0.2^{\circ}$, $9.2^{\circ}\pm 0.2^{\circ}$, $13.8^{\circ}\pm 0.2^{\circ}$ and $11.0^{\circ}\pm 0.2^{\circ}$, using CuK α radiation; preferably, the X-ray powder diffraction pattern of Form CSII comprises characteristic peaks at 2theta values of $7.9^{\circ}\pm 0.2^{\circ}$, $9.2^{\circ}\pm 0.2^{\circ}$, $13.8^{\circ}\pm 0.2^{\circ}$ and $11.0^{\circ}\pm 0.2^{\circ}$ using CuK α radiation.

[0016] In other aspect provide herein, the X-ray powder diffraction pattern of Form CSII comprises three or four or five or six or seven or eight or nine or ten characteristic peaks at 2theta values of $10.3^{\circ}\pm 0.2^{\circ}$, $12.1^{\circ}\pm 0.2^{\circ}$, $15.4^{\circ}\pm 0.2^{\circ}$, $7.9^{\circ}\pm 0.2^{\circ}$, $9.2^{\circ}\pm 0.2^{\circ}$, $13.8^{\circ}\pm 0.2^{\circ}$, $11.0^{\circ}\pm 0.2^{\circ}$, $11.5^{\circ}\pm 0.2^{\circ}$, $17.4^{\circ}\pm 0.2^{\circ}$ and $21.8^{\circ}\pm 0.2^{\circ}$, using CuK α radiation.

[0017] Without any limitation being implied, the X-ray powder diffraction pattern of Form CSII is substantially as depicted in FIG. 1 or FIG. 2 using CuK α radiation.

[0018] Without any limitation being implied, Form CSII is an anhydrate.

[0019] According to the objective of the present disclosure, a process for preparing Form CSII is also provided. The process comprises:

[0020] Adding Compound I dihydrochloride into chloroform, stirring, separating and drying to obtain Form CSII.

[0021] The stirring temperatures is preferably 50°C. ; the stirring is preferably more than 24 hours.

[0022] According to the objective of the present disclosure, the present disclosure also provides the use of Form CSII for preparing other crystalline forms, salts or cocrystals of Compound I.

[0023] According to the objective of the present disclosure, a pharmaceutical composition is provided, the pharmaceutical composition comprises a therapeutically effective amount of Form CSII and pharmaceutically acceptable excipients.

[0024] According to the objective of the present disclosure, Form CSII of present disclosure can be used for preparing estrogen receptor depressant drugs.

[0025] According to the objective of the present disclosure, Form CSII of present disclosure can be used for preparing drugs treating breast cancer with ER+, HER2-, ESR1 mutations.

ADVANTAGES AND TECHNICAL PROBLEMS SOLVED BY THE PRESENT DISCLOSURE

[0026] The technical problem solved by the present disclosure is to provide a novel crystalline form different from the crystalline forms in the prior arts. On the basis of excellent stability, the crystalline form also has higher solubility, better flowability, better compressibility, lower adhesiveness, better purification effect, no agglomeration and more uniform particle size distribution compared with the prior art Form 1B, which solves the problems existing in the prior arts.

[0027] Form CSII of the present disclosure has the following unexpected beneficial effects:

[0028] (1) Compared with the prior art Form 1B, Form CSII of the present disclosure has better solubility. Especially in FeSSIF, the solubility is more than twice that of the prior art Form 1B.

[0029] Elacestrant is a poorly water-soluble drug and belongs to BCS class IV. It is a drug with low solubility and low permeability, which are difficult to release and absorb in vivo and have low bioavailability. Higher solubility of Form CSII drug substance provided by the present disclosure is beneficial to improve drug's in vivo absorption and bioavailability. In addition, drug dose reduction without affecting efficacy is possible due to higher solubility, thereby reducing the drug's side effects and improving drug safety.

[0030] (2) Compared with the prior art Form 1B, Form CSII of the present disclosure has better purification effect. The purity is significantly increased after the raw material is converted into Form CSII of the present disclosure. In a specific embodiment of the present disclosure, the purity of the raw material used is 99.14%. The purity of Form CSII made from the raw material is 99.80%. The purity is increased by 0.60%. Especially for impurities with RRT of 0.84, 1.14 and

1.43, Form CSII has significant purification ability. Chemical purity is of great significance for ensuring drug efficacy, safety and preventing the occurrence of adverse effects. As Form CSII of the present disclosure has good purification ability and is excellent in removing impurities in the crystallization process. Thus drug substances with high purity can be obtained through crystallization, which effectively overcome the disadvantages of poor stability, poor efficacy and high toxicity caused by the low purity drug substances.

[0031] (3) Compared with the prior art Form 1B, Form CSII of the present disclosure has no agglomeration, and more uniform of particle size distribution. A uniform particle size helps to reduce solvent enrichment, improve product purity, reduce product residue, ensure the uniformity of drug formulation content, and reduce the variability of in vitro dissolution.

[0032] (4) Compared with the prior art Form 1B, Form CSII of the present disclosure better flowability. Flowability evaluation results indicate that the flowability of Form CSI is remarkably better than that of prior art forms. Better flowability can prevent clogging of production equipment and increase manufacturing efficiency. Better flowability of Form CSI ensures the content uniformity of the drug product, reduces the weight variation of the drug product and improves product quality.

[0033] (5) Compared with the prior art Form 1B, Form CSII of the present disclosure has better compressibility. Failure in hardness/friability test and tablet crack issue can be avoided due to better compressibility, making the preparation process more reliable, improving product appearance, promoting product quality and production efficiency.

[0034] (6) Compared with the prior art Form 1B, Form CSII of the present disclosure shows lower adhesiveness. Adhesiveness evaluation results indicate that adhesion quantity of Form CSII is remarkably lower than that of prior art forms. Low adhesiveness of Form CSII can reduce the agglomeration of drug substance and effectively improve the adhesion to roller and tooling during dry-granulation and compression process. It is conducive to the dispersion of drug substance with excipients and improving the blend uniformity of the mixing of materials, which ultimately improves product quality.

[0035] (7) Form CSII of the present disclosure has good physical stability under mechanical force. The crystalline form of Form CSII doesn't change after grinding and formulation process. Grinding and pulverization are often required in the drug manufacturing process. Good physical stability of the drug substance can reduce the risk of crystallinity decrease and crystal transformation during the drug production process. Form CSII has good physical stability under different pressures, which is beneficial to keep crystalline form unchanged during tableting process.

[0036] (8) From CSII drug substance of the present disclosure has good stability itself and in drug product. Crystalline state of Form CSII drug substance doesn't change for at least 6 months when stored under the condition of 25° C./60% RH. The chemical purity remains substantially unchanged during storage. After Form CSII is mixed with the excipients to form a drug product and stored under the condition of 25° C./60% RH, crystalline state of Form CSII drug product doesn't change for at least 3 months. These results show that From CSII drug substance of the present disclosure has good stability under long term condition both itself and in drug product, which is beneficial to the drug storage.

[0037] Meanwhile, crystalline state of Form CSII drug substance doesn't change for at least 6 months when stored under the condition of 40° C./75% RH. The chemical purity remains substantially unchanged during storage. After Form CSII is mixed with the excipients to form a drug product and stored under the condition of 40° C./75% RH, crystalline state of Form CSII drug product doesn't change for at least 1 month. These results show that Form CSII drug substance has good stability under accelerated and stress conditions both itself and in drug product. Therefore, good stability under accelerated and stress conditions is of great importance to the drug development.

[0038] Good physical and chemical stability of drug substance ensure that no crystal

transformation or impurities is generated during production and storage. Form CSI has good physical and chemical stability, ensuring consistent and controllable quality of the drug substance and drug product, minimizing quality change, bioavailability change and toxicity due to crystal transformation or impurity generation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. 1 shows an XRPD pattern of Form CSII;

[0040] FIG. 2 shows an XRPD pattern of Form CSII;

[0041] FIG. 3 shows a TGA curve of Form CSII;

[0042] FIG. 4 shows an XRPD pattern overlay of Form CSII before and after storage with different conditions (from top to bottom: initial, 25° C./60% RH for 6 months sealed with desiccant and antioxidant; 40° C./75% RH for 6 months sealed with desiccant and antioxidant);

[0043] FIG. 5 shows an XRPD pattern overlay of Form CSII before and after ball milling;

[0044] FIG. 6 shows an XRPD pattern overlay of Form CSII before and after formulation process (from top to bottom: blank mixture, after formulation process, Form CSII);

[0045] FIG. 7 shows an XRPD pattern overlay of Form CSII drug product before and after storage with different conditions (from top to bottom: initial, 25° C./60% RH for 3 months sealed with 1 g desiccant and 1 g antioxidant; 40° C./75% RH for 1 months sealed with 1 g desiccant and 1 g antioxidant).

DETAILED DESCRIPTION

[0046] The present disclosure is further illustrated by the following examples which describe the preparation and use of the crystalline forms of the present disclosure in detail. It is obvious to those skilled in the art that changes in the materials and methods can be accomplished without departing from the scope of the present disclosure.

[0047] The abbreviations used in the present disclosure are explained as follows: [0048] RH: Relative Humidity [0049] XRPD: X-ray Powder Diffraction [0050] TGA: Thermo Gravimetric Analysis [0051] HPLC: High Performance Liquid Chromatography [0052] HDPE: High Density Polyethylene [0053] FeSSIF: Fed-State Simulated Intestinal Fluid

Instruments and Methods Used for Data Collection

[0054] X-ray powder diffraction patterns in the present disclosure were acquired by a Bruker X-ray powder diffractometer. The parameters of the X-ray powder diffraction method of the present disclosure are as follows: [0055] X-Ray source: Cu, K α [0056] K α 1 (Å): 1.54060; K α 2 (Å): 1.54439 [0057] K α 2/K α 1 intensity ratio: 0.50

[0058] Thermo gravimetric analysis (TGA) data in the present disclosure were acquired by a TA Q500. The parameters of the TGA method of the present disclosure are as follows: [0059] Heating rate: 10° C./min [0060] Purge gas: nitrogen

[0061] The particle size distribution data in the present disclosure were acquired by an Mastersizer 3000 laser particle size analyzer of Malvern. The test was carried out in wet mode, using a Hydro MV dispersion device, and the dispersant was Isopar G. The parameters are as follows:

TABLE-US-00001 Size distribution: Volume Scattering model: Mie Dispersant name: Isopar G Fluid refractive index: 1.420 Number of measurements: 3 times Measurement duration: 10 s Absorption index: 0.100 Particle refractive index: 1.520 Particle type: Irregularity Analysis model: General Stirrer speed: 2000 rpm N/A

[0062] The parameters for related substance and Compound I detection in the present disclosure are shown in Table 1.

TABLE-US-00002 TABLE 1 HPLC Waters ACQUITY UPLC H-Class plus with PDA detector Column Waters ACQUITY UPLC BEH Shield RP18, 2.1 mm × 50 mm, 1.7 μ m Mobile Phase A:

0.1% H.sub.3PO.sub.4 in H.sub.2O (pH 2.5, TEA) B: ACN:MeOH = 800:200 (v/v) Gradient Time (min) % B 0.0 10 9.0 40 12.0 80 16.0 80 16.1 10 18.0 10 Flow Rate 0.5 mL/min Injection Volume 1 μ L Detector Wavelength UV at 240 nm Column Temperature 40° C. Sample Temperature Room Temperature Diluent Acetonitrile:H.sub.2O = 50:50 (v/v)

[0063] The parameters for kinetic solubility in the present disclosure are shown in Table 2.

TABLE-US-00003 TABLE 2 HPLC Waters ACQUITY UPLC H-Class plus with PDA director Column Waters ACQUITY UPLC BEH Shield RP18, 2.1 mm $\times$ 50 mm, 1.7 μ m Mobile Phase A: 0.1% H.sub.3PO.sub.4 in H.sub.2O (pH 2.5, TEA) B: ACN:MeOH = 800:200 (v/v) Gradient Time (min) % B 0.0 10 9.0 40 12.0 80 13.0 80 13.1 10 15.0 10 Flow Rate 0.5 mL/min Injection Volume 1 μ L Detector Wavelength 215 nm Column Temperature 40° C. Sample Temperature Room Temperature Diluent Acetonitrile:H.sub.2O = 50:50 (v/v)

[0064] The parameters for the content of chloride ion in the present disclosure are shown in Table 3.

TABLE-US-00004 TABLE 3 IC Thermo Fisher Dionex Aquion Column Thermo Dionex IonPac AS22, 4 $\times$ 250 mm, 6.0 μ m Eluent 4.5 mM Na.sub.2CO.sub.3/1.4 mM NaHCO.sub.3 Injection Volume 25 μ L Flow Rate 1.0 mL/min Conductivity Cell 35° C. Temperature Column 30° C. Temperature: Current 31 mA Run time 8-10 min

[0065] In the present disclosure, the “stirring” is accomplished by using a conventional method in the field such as magnetic stirring or mechanical stirring and the stirring speed is 50 to 1800 r/min. Preferably the magnetic stirring speed is 300 to 900 r/min and mechanical stirring speed is 100 to 300 r/min.

[0066] The “separation” is accomplished by using a conventional method in the field such as centrifugation or filtration. The operation of “centrifugation” is as follows: the sample to be separated is placed into the centrifuge tube, and then centrifuged at a rate of 10000 r/min until the solid all sinks to the bottom of the tube.

[0067] The “drying” is accomplished by using a conventional method in the field such as vacuum drying, blast drying or free-air drying. The drying temperature can be room temperature or higher. Preferably the drying temperature is from room temperature to about 60° C., or to 50° C., or to 40° C. The drying time can be 2 to 48 hours, or overnight. Drying is accomplished in a fume hood, forced air convection oven or vacuum oven.

[0068] The “characteristic peak” refers to a representative diffraction peak used to distinguish crystals, which usually can have a deviation of $\pm 0.2^\circ$ using CuK α radiation.

[0069] In the present disclosure, “crystal” or “crystalline form” refers to the crystal or the crystalline form being identified by the X-ray diffraction pattern shown herein. Those skilled in the art are able to understand that the X-ray powder diffraction pattern depend on the instrument conditions, the sample preparation and the purity of samples. The relative intensity of the diffraction peaks in the X-ray diffraction pattern may also vary with the experimental conditions; therefore, the order of the diffraction peak intensities cannot be regarded as the sole or decisive factor. In fact, the relative intensity of the diffraction peaks in the X-ray powder diffraction pattern is related to the preferred orientation of the crystals, and the diffraction peak intensities shown herein are illustrative and identical diffraction peak intensities are not required. Thus, it will be understood by those skilled in the art that a crystalline form of the present disclosure is not necessarily to have exactly the same X-ray diffraction pattern of the example shown herein. Any crystalline forms whose X-ray diffraction patterns have the same or similar characteristic peaks should be within the scope of the present disclosure. Those skilled in the art can compare the patterns shown in the present disclosure with that of an unknown crystalline form in order to identify whether these two groups of patterns reflect the same or different crystalline forms.

[0070] In some embodiments, Form CSII of the present disclosure is pure and substantially free of any other crystalline forms. In the present disclosure, the term “substantially free” when used to describe a novel crystalline form, it means that the content of other crystalline forms in the novel

crystalline form is less than 20% (w/w), specifically less than 10% (w/w), more specifically less than 5% (w/w) and furthermore specifically less than 1% (w/w).

[0071] In the present disclosure, the term “about” when referring to a measurable value such as weight, time, temperature, and the like, is meant to encompass variations of +10%, ±5%, ±1%, ±0.5%, or even ±0.1% of the specified amount.

[0072] Unless otherwise specified, the following examples were conducted at room temperature and environment humidity. The “room temperature” is not a specific temperature, but a temperature range of 10-30° C. The “environment humidity” is not a specific humidity, but a humidity range of 10% RH-90% RH, 20% RH-80% RH or 30% RH-70% RH.

[0073] According to the present disclosure, Compound I and/or its salt used as raw materials include, but are not limited to solid (crystalline and amorphous), semisolid, wax, oil, liquid form or solution. Preferably, Compound I and/or its salt used as the raw material is a solid.

[0074] Raw materials of Compound I and/or a salt thereof used in the following examples were prepared by known methods in the prior art, for example, the method disclosed in WO2018129419A1.

Example 1: Preparation of Form CSII

[0075] 10.6 mg of Compound I dihydrochloride was weighed into a glass vial, and 0.1 mL of chloroform was added thereafter. The suspension was stirred at 50° C. for 21 days, then transferred to -20° C. and kept for 6 days. The solid was separated and dried under vacuum overnight to obtain dry solid.

[0076] The dry solid was confirmed to be Form CSII by XRPD. The XRPD pattern is substantially as depicted in FIG. 1, and the XRPD data are listed in Table 4.

TABLE-US-00005 TABLE 4 2θ d-spacing Intensity (°) (Å) (%) 7.57 11.68 11.90 7.91 11.17 23.43 9.21 9.60 39.27 10.33 8.56 100.00 11.01 8.04 50.43 11.46 7.72 53.27 12.09 7.32 89.37 12.83 6.90 13.06 13.76 6.43 52.55 15.39 5.76 58.61 15.91 5.57 18.80 17.42 5.09 72.23 18.02 4.92 39.62 18.44 4.81 58.26 18.74 4.73 48.77 19.26 4.61 33.06 20.29 4.38 26.21 20.74 4.28 33.38 21.80 4.08 91.98 23.22 3.83 32.66 24.57 3.62 27.99 26.05 3.42 27.24 26.77 3.33 20.99 28.11 3.17 50.24 28.98 3.08 24.03 30.36 2.94 10.83 31.69 2.82 17.08 32.94 2.72 4.01

[0077] The dry solid was confirmed to be Form CSII by XRPD. The XRPD pattern is substantially as depicted in FIG. 1, and the XRPD data are listed in Table 4.

Example 2: Preparation of Form CSII

[0078] 99.7 mg of Compound I dihydrochloride was weighed into a glass vial, and 1 mL of chloroform was added thereafter. An appropriate amount of Form CSII seeds was added. The suspension was stirred for 3 days at 50° C., then transferred to room temperature and dried under vacuum for 2 days to obtain dry solid.

[0079] The dry solid was confirmed to be Form CSII by XRPD. The XRPD pattern is substantially as depicted in FIG. 2, and the XRPD data are listed in Table 5.

TABLE-US-00006 TABLE 5 2θ d-spacing Intensity (°) (Å) (%) 7.57 11.68 7.41 7.91 11.17 23.56 9.24 9.57 35.19 10.33 8.56 98.60 11.03 8.02 39.29 11.48 7.71 42.26 12.09 7.32 79.28 12.84 6.89 8.55 13.36 6.63 15.41 13.80 6.42 52.22 15.42 5.75 41.78 15.93 5.57 18.59 17.45 5.08 68.10 18.46 4.81 48.92 18.75 4.73 41.93 19.27 4.61 28.72 20.33 4.37 23.38 20.75 4.28 30.90 21.78 4.08 100.00 22.24 4.00 64.39 23.11 3.85 33.62 23.81 3.74 24.86 24.64 3.61 28.93 26.09 3.42 26.45 28.14 3.17 54.52 29.01 3.08 23.46 30.41 2.94 9.56 31.68 2.82 21.10 33.04 2.71 5.00

Example 3 TGA Curve of Form CSII

[0080] The TGA curve of Form CSII is substantially as depicted in FIG. 3, and it shows Form CSII almost has no weight loss when heated to 100° C.

Example 4 Content of Chloride Ion of Form CSII

[0081] The content of chloride ion and free base of Form CSII of the present disclosure was tested, and the molar ratio was calculated. The results are shown in Table 6. The results indicate that the molar ratio of Form CSII of the present disclosure is 2.0 and Form CSII is a dihydrochloride.

TABLE-US-00007 TABLE 6 Molar ratio (chloride Form ion:free base) Form CSII 2.03 1.98

Example 5 Kinetic Solubility of Form CSII

[0082] When solubility test is used to predict the in vivo performance of a drug, it is critical to simulate in vivo conditions as closely as possible. For oral medication, Fed-State Simulated Intestinal Fluid (FeSSIF) can be used to simulate the condition in vivo and predict the effects of feeding, thus solubility in such mediums is closer to that in vivo.

[0083] 10 mg of Form CSII and 10 mg of prior art Form 1B were suspended into 1 mL of FeSSIF to get suspensions. After equilibrated at 37° C. for 30 min, the suspensions were filtrated to get saturated solutions. The concentrations (mg/mL) of Compound I of the saturated solutions were measured by HPLC. The results are listed in Table 7. The results show that the solubility of Form CSII in FeSSIF is higher than that of Form 1B, and the solubility of Form CSII is 2.2 times that of Form 1B.

TABLE-US-00008 TABLE 7 Solubility (mg/mL) Prior art Form Medium Form 1B CSII FeSSIF
1.17 2.59

Example 6 Purification Ability of Form CSII

[0084] Form CSII and prior art Form 1B were prepared with the same starting material. HPLC was applied to test the chemical purity of starting material, Form CSII and prior art Form 1B. The results are listed in Table 8. The results show that chemical purity of Form CSII is higher than that of prior art Form 1B when the same starting material was used, especially for impurities with RRT (relative retention time) of 0.84, 1.14 and 1.43. Form CSII has better purification ability than prior art Form 1B.

TABLE-US-00009 TABLE 8 Purity RRT = RRT = RRT = Form Purity increment 0.84 1.14 1.43
Starting material 99.20% — 0.04% 0.08% 0.11% Form CSII 99.80% 0.60% ND ND ND Prior art
Form 1B 99.67% 0.47% 0.05% 0.05% 0.04% ND: Not detected

Example 7 Morphology of Form CSII

[0085] An appropriate amount of Form CSII of the present disclosure was placed on a glass slide and dispersed with vacuum pump oil. It was covered with a cover glass slide and shoot at a magnification of 200 times. The results indicate that Form CSII has no agglomeration, and it has larger particles with a maximum particle size of 30-40 μm.

Example 8 Particle Size Distribution of Form CSII

[0086] An appropriate amount of Form CSII of the present disclosure and prior art Form 1B was tested particle size distribution. The average particle diameter calculated by volume, the diameter at which 10% mass is comprised of smaller particles (D10), the diameter at which 50% mass is comprised of smaller particles (D50) and the diameter at which 90% mass is comprised of smaller particles (D90) were obtained in particle size distribution test. The results are shown in Table 9. The results show that the particle size distribution of Form CSII is uniform, which is superior to that of prior art Form 1B.

TABLE-US-00010 TABLE 9 Form MV (μm) D10 (μm) D50 (μm) D90 (μm) Form CSII 15.8 2.11
14.9 28.3 Prior art Form 1B 350 1.64 286 839

Example 9 Flowability of Form CSII

[0087] Compressibility index is usually utilized to evaluate the flowability of powder or granules during the drug product process. Compressibility index test method is as follows: a certain amount of powder was added into a measuring cylinder and bulk volume was recorded. Then the powder was tapped to make it in the tightest state and the tapped volume was recorded. The bulk density (p0), tapped density (pf) were calculated and compressibility index was calculated according to $c = (pf - p0) / pf$. According to the criteria of flowability in USP General Chapters: <1174> Powder Flow, the flowability results are shown in Table 10. The results indicate that flowability of Form CSII is remarkably superior to that of the prior art Form 1B.

TABLE-US-00011 TABLE 10 Bulk Tapped density density Compressibility Form (g/m) (g/mL)
index Flowability Prior art Form 1B 0.198 0.248 20 Fair Form CSII 0.237 0.266 11 Good

Example 10 Compressibility of CSII

[0088] An ENERPAC manual tablet press was used for compression. 60 mg of Form CSII and prior art Form 1B were weighed and added into the dies of a $\phi 6$ mm round tooling, compressed at 5 KN manually, then stored at room temperature for 24 h until elastic recovery is complete, hardness (H) was tested with an intelligent tablet hardness tester. Diameter (D) and thickness (L) were tested with a caliper. Tensile strength of the powder was calculated with the following formula: $T=2H/TTDL$. Under a certain force, the greater the tensile strength, the better the compressibility. The results are presented in Table 11. The results indicate that Form CSII has better compressibility compared with prior art Form 1B.

TABLE-US-00012 TABLE 11 Tensile Hardness Diameter Thickness strength Form (kgf) (mm) (mm) (MPa) Prior art Form 1B 9.3 6.06 2.21 0.44 Form CSII 18.4 6.04 2.00 0.97

Example 11 Adhesiveness of Form CSII

[0089] 30 mg of Form CSII and 30 mg of prior art Form 1B were weighed and then added into the dies of $\Phi 8$ mm round tooling, compressed at 10 KN by ENERPAC manual tablet press and held for 30 s. The punch was weighed and amount of material sticking to the punch was calculated. The compression was repeated twice and the average amount of material sticking to the punch during the compression were calculated. Detailed experimental results are shown in Table 12. Test results indicate that the adhesiveness of Form CSII is lower than that of prior art Form 1B.

TABLE-US-00013 TABLE 12 Average amount Form (mg) Prior art Form 1B 0.59 Form CSII 0.21

Example 12 Stability of Form CSII

[0090] An appropriate amount of Form CSII of the present disclosure were stored under different sealed conditions of 25° C./60% RH and 40° C./75% RH with desiccant and antioxidant. Chemical purity and crystalline form were checked by HPLC and XRPD, respectively. The results are shown in Table 13, and the XRPD overlay is shown in FIG. 5. The results show that Form CSII kept stable for at least 6 months at sealed conditions of 25° C./60% RH and 40° C./75% RH with desiccant and antioxidant. Form CSII has good stability under both long-term and accelerated conditions.

TABLE-US-00014 TABLE 13 Initial Condition Package Time Form Purity Form CSII Initial — — Form CSII 99.59% 25° C. / Sealed package 6 months Form CSII 99.60% 60% RH with desiccant 40° C. / and antioxidant 6 months Form CSII 99.58% 75% RH

Example 13 Stability of Form CSII Upon Mechanical Force

[0091] Approximately 20 mg of Form CSII was added into the dies of a $\phi 6$ mm round tooling, compressed into tablets under different pressures. Crystalline forms before and after tableting were checked by XRPD. The test results are shown in Table 14. The results show that Form CSII has better stability under different pressures.

TABLE-US-00015 TABLE 14 Before Solid form after tableting Pressure tableting Form CSII 5 kN Form CSII 10 kN Form CSII 15 kN Form CSII

Example 14 Preparation of Form CSII Drug Product

[0092] Form CSII of the present disclosure was made into capsules using the formulation in Table 15 and the preparation process in Table 16. The XRPD patterns were collected before and after the formulation process. The XRPD overlay is shown in FIG. 6. The results indicate that Form CSII remains stable before and after the formulation process.

TABLE-US-00016 TABLE 15 No. Component mg/unit % (w/w) 1 Compound I dihydrochloride (Form CSII) 23.2* 23.2* 2 Microcrystalline cellulose 51.4 51.4 3 Lactose 19.4 19.4 4 Sodium carboxymethyl starch 5 5 5 Magnesium stearate 0.5 0.5 6 Colloidal silicon dioxide 0.5 0.5 Total 100 100 Note: The mass of 23.2 mg of Compound I dihydrochloride corresponds to 20 mg of Compound I free base.

TABLE-US-00017 TABLE 16 Stage Procedure Pre-blending According to the formulation, materials No. 1-6 were weighed into an LDPE bags and blended for 2 mins. Simulation The mixture was pressed by a single punch manual of dry tablet press (type: ENERPAC, die: Φ 20 mm round, granulation pressure: 5 ± 1 KN). The flakes were pulverized and sieved through a 20-mesh

sieve. Tableting The total mixed powder (approximately 100 mg) was added into a T 8 × 4 punch die and compressed at 5 KN to obtain a tablet. Package A tablet was sealed in 35 cc HDPE bottle with 1 g of desiccant and 1 g of antioxidant.

Example 15 Stability of Form CSII Drug Product

[0093] The Form CSII drug product was stored under sealed conditions of 25° C./60% RH and 40° C./75% RH with 1 g of desiccant and 1 g of antioxidant, and the crystalline forms before and after storage were tested to evaluate its stability. The results are shown in Table 17 and the XRPD comparison graph is shown in FIG. 7. The results show that Form CSII drug product kept stable for at least 3 months at 25° C./60% RH and at least 1 month at 40° C./75% RH conditions, which indicate Form CSII drug product has good stability under both long-term and accelerated conditions.

TABLE-US-00018 TABLE 17 Initial Condition Package Time Form Purity Form CSII Initial — —
Form CSII 99.64% 25° C. / Sealed package 3 months Form CSII 99.60% 60% RH with desiccant
40° C. / and antioxidant 1 months Form CSII 99.60% 75% RH

[0094] The examples described above are only for illustrating the technical concepts and features of the present disclosure and intended to make those skilled in the art being able to understand the present disclosure and thereby implement it and should not be concluded to limit the protective scope of this disclosure. Any equivalent variations or modifications according to the spirit of the present disclosure should be covered by the protective scope of the present disclosure.

Claims

1. A crystalline form of Compound I dihydrochloride, wherein the X-ray powder diffraction pattern comprises characteristic peaks at 2theta values of $10.3^{\circ} \pm 0.2^{\circ}$, $12.1^{\circ} \pm 0.2^{\circ}$ and $15.4^{\circ} \pm 0.2^{\circ}$ using CuK α radiation ##STR00003##
 2. The crystalline form of Compound I dihydrochloride according to claim 1, wherein the X-ray powder diffraction pattern comprises at least one characteristic peaks at 2theta values of $7.9^{\circ} \pm 0.2^{\circ}$, $9.2^{\circ} \pm 0.2^{\circ}$, $13.8^{\circ} \pm 0.2^{\circ}$ and $11.0^{\circ} \pm 0.2^{\circ}$ using CuK α radiation.
 3. The crystalline form of Compound I dihydrochloride according to claim 1, wherein the X-ray powder diffraction pattern of the crystalline form is substantially as depicted in FIG. 1 or FIG. 2 using CuK α radiation.
 4. The crystalline form of Compound I dihydrochloride according to claim 1, wherein the crystalline form is an anhydrate.
 5. A pharmaceutical composition, wherein the pharmaceutical composition comprises a therapeutically effective amount of crystalline form of Compound I dihydrochloride according to claim 1, and pharmaceutically acceptable excipients.
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